PS#14 - The normal distribution! – KEY

1. Draw a graph of $p(x)$.
2. (Shade in the region of integration.)
3. (Determine bounds of integration.) From $a = -1$ to $b = 1$.
4. (Not integrable.)
5. Write down the Taylor series for $f(u) = e^u$.

$$e^u = \sum_{k=0}^{\infty} \frac{1}{k!} u^k = 1 + \frac{u}{1!} + \frac{u^2}{2!} + \frac{u^3}{3!} + \dots$$

6. Use the Taylor series for e^u to write down the Taylor series for $p(x)$.

$$T_p(x) = \frac{1}{\sqrt{2\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{k! \cdot 2^k} = \frac{1}{\sqrt{2\pi}} \left(1 - \frac{x^2}{1! \cdot 2} + \frac{x^4}{2! \cdot 4} - \frac{x^6}{3! \cdot 8} + \frac{x^8}{4! \cdot 16} \dots\right)$$

7. Use the Taylor series for $p(x)$ to write down a Taylor series for its antiderivative $P(x)$.

$$T_P(x) = \frac{1}{\sqrt{2\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{k! \cdot 2^k \cdot (2k+1)} = \frac{1}{\sqrt{2\pi}} \left(x - \frac{x^3}{1! \cdot 2 \cdot 3} + \frac{x^5}{2! \cdot 4 \cdot 5} - \frac{x^7}{3! \cdot 8 \cdot 7} + \frac{x^9}{4! \cdot 16 \cdot 9} \dots\right)$$

8. Use the first four nonzero terms of the Taylor series for $P(x)$ to approximate $P(b)$.

$$T_P(1) = \frac{1}{\sqrt{2\pi}} \left(1 - \frac{1^3}{1! \cdot 2 \cdot 3} + \frac{1^5}{2! \cdot 4 \cdot 5} - \frac{1^7}{3! \cdot 8 \cdot 7}\right) = \frac{1}{\sqrt{2\pi}} \cdot \frac{479}{560} \approx 0.34124.$$

9. Use the first four nonzero terms of the Taylor series for $P(x)$ to approximate $P(a)$.

$$T_P(-1) = \frac{1}{\sqrt{2\pi}} \left(-1 + \frac{1^3}{1! \cdot 2 \cdot 3} - \frac{1^5}{2! \cdot 4 \cdot 5} + \frac{1^7}{3! \cdot 8 \cdot 7}\right) = -\frac{1}{\sqrt{2\pi}} \cdot \frac{479}{560} \approx -0.34124.$$

10. Find $P(b) - P(a)$. (Your answer should be close to 68%.)

$$P(1) - P(-1) \approx \frac{1}{\sqrt{2\pi}} \cdot \frac{479}{560} - \left(-\frac{1}{\sqrt{2\pi}} \cdot \frac{479}{560}\right) \approx 0.34124 + 0.34124 = 0.68248.$$

11. How many terms would be necessary to ensure that your approximation to $P(b) - P(a)$ is accurate to four decimal places?

k	1	2	3	4	5	6
$\lvert a_k \rvert$	0.1329808	0.0199471	0.0023747	0.0002309	0.0000189	0.0000013

So the first term that's smaller than 5×10^{-5} is $|a_5|$, which means we need $k = 4$, which means we need five terms. (Remember, k starts counting at 0.)